

의약품공학(이) (LS312-01)

학기	2026년 1학기
담당교수	예경무
이수구분	교선포필수
학점	3.0
강의시간	화09:00-12:00(E4 - 113)
강의실	
소속	기초학부

교과개요

Pharmaceuticals are largely classified into chemical drugs based on synthetic chemicals and bio-pharmaceuticals based on biological materials. In this lecture, you will learn about the types and characteristics of representative drugs, medical applications, and development strategies. In this lecture, Professor Byung-Hoon Lee, Ph.D, an expert in the development of chemical drugs based on protein degradation and Professor Kyungmoo Yea, Ph.D, an expert in the development of bio-pharmaceuticals based on antibodies and exosomes, collaborate to cover a wide range of knowledge in the pharmaceutical field.

교과목표

The purpose of this course is to lay the foundation for professional courses. It begins with an introduction to synthetic chemicals and bio-pharmaceuticals, which are essential aspects of therapeutic development in the field of biology, and is recommended for undergraduate students who wish to build prior knowledge essential for future professional courses.

주차별 강의계획

1주차:Course introduction
2주차:Introduction of bio- pharmaceuticals
3주차:Protein therapeutics:Antibody, recombinant protein drugs
4주차:Protein library construction and screening/selection
5주차:Antibody engineering-I
6주차:Antibody engineering-II
7주차:Application of antibody engineering
8주차:Mid-term exam
9주차:Introduction of chemical drugs
10주차:History of natural products and chemical drugs
11주차:Nobel Prize awarded drugs
12주차:Drug discovery: targets
13주차:Drug discovery: screening
14주차:Drug development
15주차:Innovation drugs: targeted protein degradation
16주차:Final exam

학습윤리

Dishonest academic practices, such as cheating and plagiarism, are unacceptable. Engaging in such behaviors will lead to receiving no points on the exams and will be addressed as per the university's disciplinary standards.

교과정책(방침)

Cell phones should be turned off or set to silent mode during class.

No texting or browsing on phones during class.

Laptops/tablets should only be used for course-related activities.